

CR-BIS Data Acquisition User Guide

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1. Introduction

This guide describes the use of the **CR-BIS (Continuous Recording Beam-Image Shift)** data acquisition script for **SerialEM**. Users can customize parameters as needed and perform multiple acquisition modes, including conventional BIS and CR- BIS. The following sections focus on the operation of the CR-BIS mode and its required pre-calibration steps.

Part 1: Pre-calibration

1.1 Measuring Camera Delay Curves

Different cryo-EM camera models exhibit distinct delay characteristics. It is recommended to measure the delay curve in advance to quantify **the theoretical acceleration** achievable with CR-BIS.

Use the script **Report_Clock_script.txt**. Set a series of exposure times and perform five repeated measurements for each setting. Plot the relationship between the set exposure time and actual recorded time to obtain the intercept and slope parameters.

```
start 0.3 s
3 frames were saved to X:\userdata\20250520_yq_test\K2_0006.tif
2.53 seconds elapsed time
3 frames were saved to X:\userdata\20250520_yq_test\K2_0007.tif
2.53 seconds elapsed time
3 frames were saved to X:\userdata\20250520_yq_test\K2_0008.tif
2.50 seconds elapsed time
3 frames were saved to X:\userdata\20250520_yq_test\K2_0009.tif
2.63 seconds elapsed time
3 frames were saved to X:\userdata\20250520_yq_test\K2_0010.tif
2.56 seconds elapsed time
start 0.6 s
6 frames were saved to X:\userdata\20250520_yq_test\K2_0011.tif
2.84 seconds elapsed time
6 frames were saved to X:\userdata\20250520_yq_test\K2_0012.tif
2.83 seconds elapsed time
6 frames were saved to X:\userdata\20250520_yq_test\K2_0013.tif
2.81 seconds elapsed time
6 frames were saved to X:\userdata\20250520_yq_test\K2_0014.tif
2.83 seconds elapsed time
6 frames were saved to X:\userdata\20250520_yq_test\K2_0015.tif
2.83 seconds elapsed time
start 1 s
10 frames were saved to X:\userdata\20250520_yq_test\K2_0016.tif
3.27 seconds elapsed time
10 frames were saved to X:\userdata\20250520_yq_test\K2_0017.tif
3.25 seconds elapsed time
10 frames were saved to X:\userdata\20250520_yq_test\K2_0018.tif
3.34 seconds elapsed time
10 frames were saved to X:\userdata\20250520_yq_test\K2_0019.tif
3.23 seconds elapsed time
10 frames were saved to X:\userdata\20250520_yq_test\K2_0020.tif
3.27 seconds elapsed time
```

SerialEM log output example

1.2 Determining the Formula for CR-BIS Total Exposure Time

In CR-BIS mode, the total set exposure time is defined as:

$$T_{total\ set} = (T_{set\ per\ point} + T_{blank}) * point\ number + T_{buffer} \quad (1)$$

T_{total}: The total continuous exposure time calculated for a set of multiple target points.

T_{set per point}: The exposure time for each individual target.

T_{blank}: The time required for beam blanking/unblanking and BIS repositioning. To determine this, select a few points, run the acquisition in Fast mode, split the resulting stack file, and count the number of blank frames between two successive target points; then convert this number using the frame time to obtain T_{blank}.

T_{buffer}: A safety margin added to prevent T_{total} from being shorter than the actual total time needed to image all target points, which would cause the camera to stall and produce duplicate files.

Examples:

K3 camera: $T_{total\ set} = (T_{set\ per\ point} + 0.22\ s) * point\ number + 0.6\ s$

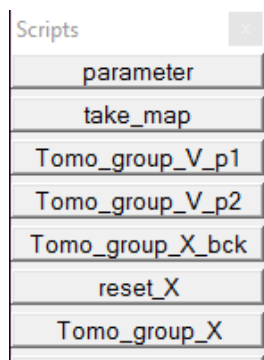
Falcon 4i camera: $T_{total\ set} = (T_{set\ per\ point} + 0.26\ s) * point\ number + 1.2\ s$

A CR-BIS acquisition collects multiple target points in a single continuous exposure. By analyzing frame rate and blank-frame data, one can estimate the beam blank/shift time and buffer duration.

Part 2: CR-BIS Data Acquisition Workflow

2.1 Overview of CR-BIS Scripts and Toolbar

The CR-BIS package includes several sub-scripts accessible through the SerialEM toolbar:



- **parameter** — Core parameter configuration script; must be executed before running other scripts.
- **take_map** — Captures low-magnification maps for target selection.
- **Tomo_group_V_p1/p2** — Performs precise alignment of target points on the low- magnification map (critical for cryo-ET).
- **reset_X** — Records key microscope parameters after setup and restores them before acquisition to ensure consistency.
- **Tomo_group_X** — Main data acquisition script for production runs.

2.2 Parameter Configuration

The parameters required for CR-BIS operation are shown in the figure below. Each colored section corresponds to a specific parameter group in the parameter script.

Script 1: parameter

```

<- Editor -> Up Script Down
Indent Fix Increase Reduce
Tab or ` to complete command
MacroName parameter

#====Tomo_group_X settings=====
#---General tomo and SPA setting-----
start_angle = 10 # first image stage tilt angle in degree
exp_start_angle = 0 # expected first image stage tilt angle in degree
min_angle = -50 # minimum stage tilt angle of the whole data, > -66
max_angle = 70 # maximum stage tilt angle of the whole data < 70
step = 3 #stage tilt step in degrees
min_defocus = -2.0 #minimum defocus value
max_defocus = -2.5 #maximum defocus value

#---microscope and camera setting-----
autoloader = 1 #set 1 for TEM with autoloader
               # 2 for not
flash_CFEG = 1 #set 1 for TEM with CFEG
               # 2 for not
camera = 1 #set 1 for using Falcon4/4i camera for data collection
            #2 for K3
            #3 for Ceta, can not be used in fast_mode

#-----output data folder setting-----
#name of folder for saving frames
folder_name @= 20250727_YtTomo_yq

# map output file path and
work_dir @= Z:\user_data\${folder_name}

# file path for saving frames, not Falcon or DE camera, e.g. K2/K3 camera only
frame_dir @= Z:\user_data\${folder_name}

```

Blue Box — Output Paths

Specify the output directory for data files.

Ensure that the map files and data output files are stored in separate directories to avoid overwriting or file management conflicts.

```
#-----data collection mode setting-----
data_mode = 1 #set 1 for dose sym tomography
               #2 for one or two direction tomography
               #3 Single particle tilt to fix angle for data collection
               #4 Single particle tilt pair
fast_mode = 1 # 1 for using fast mode, SET FAST MODE as well
               # 2 for not
accurate_position = 1 #1 for aligning each point to reference in V, 2 for not
use_focus_forZ = 2 #set 1 for focus every point to measure Z
                  #2 for not
do_eucentric_fine = 1 #set 1 for do eucentric fine each group
                    #2 for not
align_low_angle_only = 1 #1 for only align at first 4 tilts
                        #2 for align at all tilts

skip_value_limit = -100 # Value of buffer mean count to skip, lower than this value will skip collection
                       #use "ReportMeanCounts" to check value

background_stage = 1 #1 for use file_checker for stage movement when imaging
                    #2 for not

move_stage_instead_IS = 2 # 1 for only moving stage, 2 for using image shift besides stage
                        # 3 for only using image shift at high mag

#-----multi shot per hole for SPA setting-----
in_hole_distance = 0 # distance away from the selected point in micrometer, only for SPA, not for tomography and tilt pair
in_hole_num = 1 # num of image taken around the selected point, only for SPA, not for tomography and tilt pair
in_hole_start_angle = 0 # angle for the first point away from selected point, the only for SPA, not for tomography and tilt pair

#-----FAST MODE setting-----
R_delay = 0.9 # exposure time for Record mode in FAST mode
blank_delay = 0.72 # delay time in sec for the beginning of continous record mode
ObjVslSmatrix = { -0.000005 0.000049 -0.000011 -0.000050 } #do Focus/Tune -> "Calibrate Coma vs Image Shift" and cope the output ObjVslSmatrix in log
```

Red Box — CR-BIS Core Parameters

To enable CR-BIS mode, set: **fast_mode = 1**

- **R_delay** — Exposure time for each target point under CR-BIS mode.
- **blank_delay** — A microscope-dependent parameter that compensates for camera model-specific delays.
 - For K3 cameras: when switching to Continuous Mode, a short delay occurs that may cause the first target exposure to be skipped. This value should therefore be included as <blank_delay>.

- For Falcon4i cameras: there is no switching delay. However, the beam is briefly turned on at the focus position before the first target exposure, requiring an additional buffering time in the total exposure time formula.

- **ObjVsISmatrix** — Calibration matrix between objective lens deflection and image shift, used for coma and image-shift correction.

- For CR-BIS-Tomo mode, set **align_low_angle_only = 1**. This ensures the first five tilt angles use standard acquisition to obtain better alignment parameters.

Yellow Box — Fine Alignment Parameters

These parameters improve positional accuracy and are particularly important for cryo-ET.

- **accurate_position** — Optional for SPA acquisition; recommended for cryo-ET.

- **use_focus_forZ** — Measures focus at each target point at the starting tilt angle to obtain Z-height.

- Ensure the focus position differs from the actual imaging region.

- Recommended for grid-based cryo-ET in high magnification (eg. pixel size < 1.0); not suitable for FIB-lamellae.

- **do_eucentric_fine** — Performs fine eucentric height adjustment.

- Recommended for non-lamella cryo-ET samples.

- Not recommended for lamella-based cryo-ET due to potential occlusion at high tilt angles.

Purple Box — Background Stage Movement

The background_stage parameter allows pre-movement of the sample stage while CR-BIS exposures are running. This utilizes the idle camera buffering period after a long CR-BIS exposure to move the stage to the next tilt angle or focus group, saving acquisition time. To enable this feature:

1. Copy **file_checker.py** to the directory where the map files are stored.

2. Before starting acquisition, run:

python3 file_checker.py

This script monitors the exposure progress and automatically triggers stage movement once the exposure group finishes.

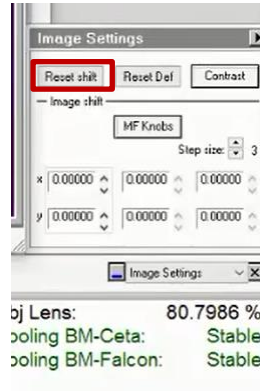
Initialization

After all parameters have been properly set, run the parameter script once. This will initialize the configuration and automatically create a data acquisition directory under the specified path for the current data collection session.

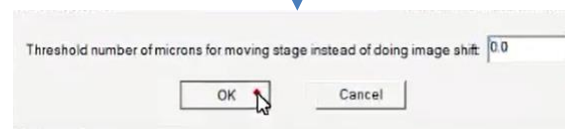
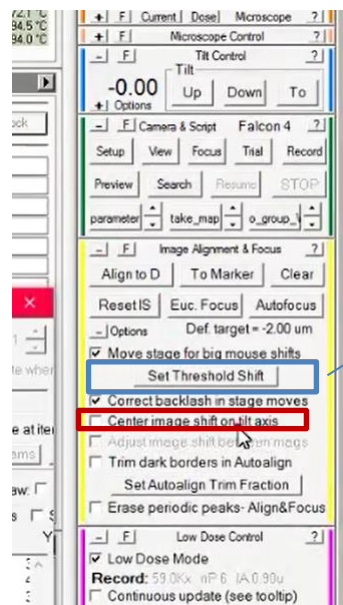
2.3 Setup Preparation

1. Verify SerialEM parameters

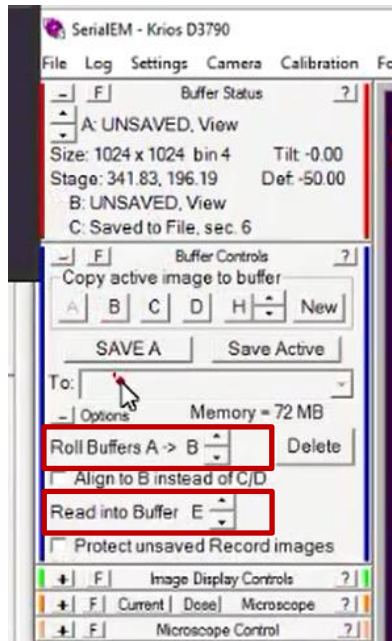
- At Record magnification, ensure *Image Settings* x and y are both 0. If not, click *Reset Shift*.



- If the shift cannot be maintained at zero, further troubleshooting is required—for example, do not check the option *Center image shift on tilt axis*, especially when the SerialEM calibration is incomplete.

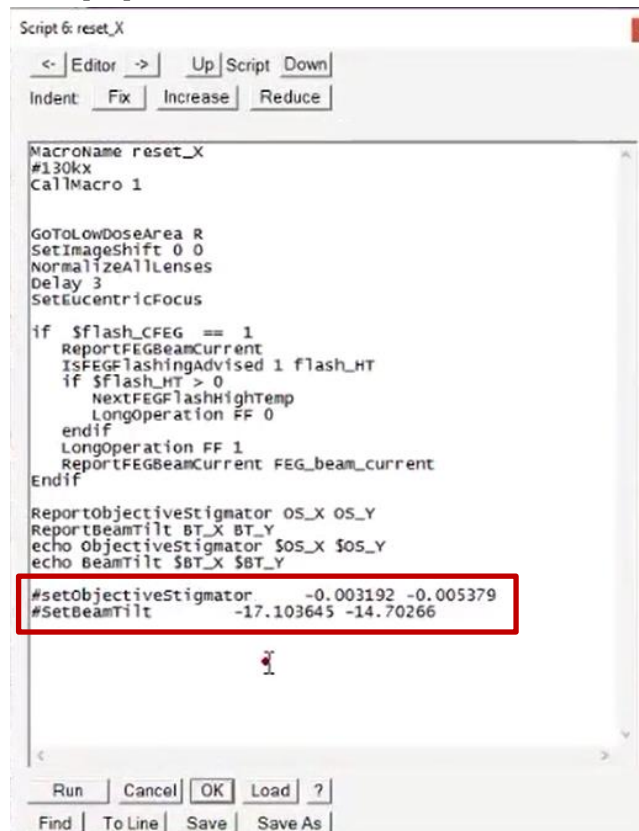


- Also verify that both values in *Set Threshold Shift* (blue box) are set to 0.0.
- Under *Buffer Controls*, set *Roll Buffers A→B* and *Read into Buffer E*; incorrect settings may cause tracking failures.



2. Reset alignment state with the *reset_X* script

- This script logs previous alignment parameters for quick recovery. Before starting alignment, Ctrl-click to open the script, **comment out the last two lines with #**, and run it once. The log file will output the current microscope parameters.



```

Log
Started a long running operation: flashing FEG
Call for flashing FEG finished successfully
ObjectiveStigmator -0.003192 -0.005379
BeamTilt -17.103645 -14.70266

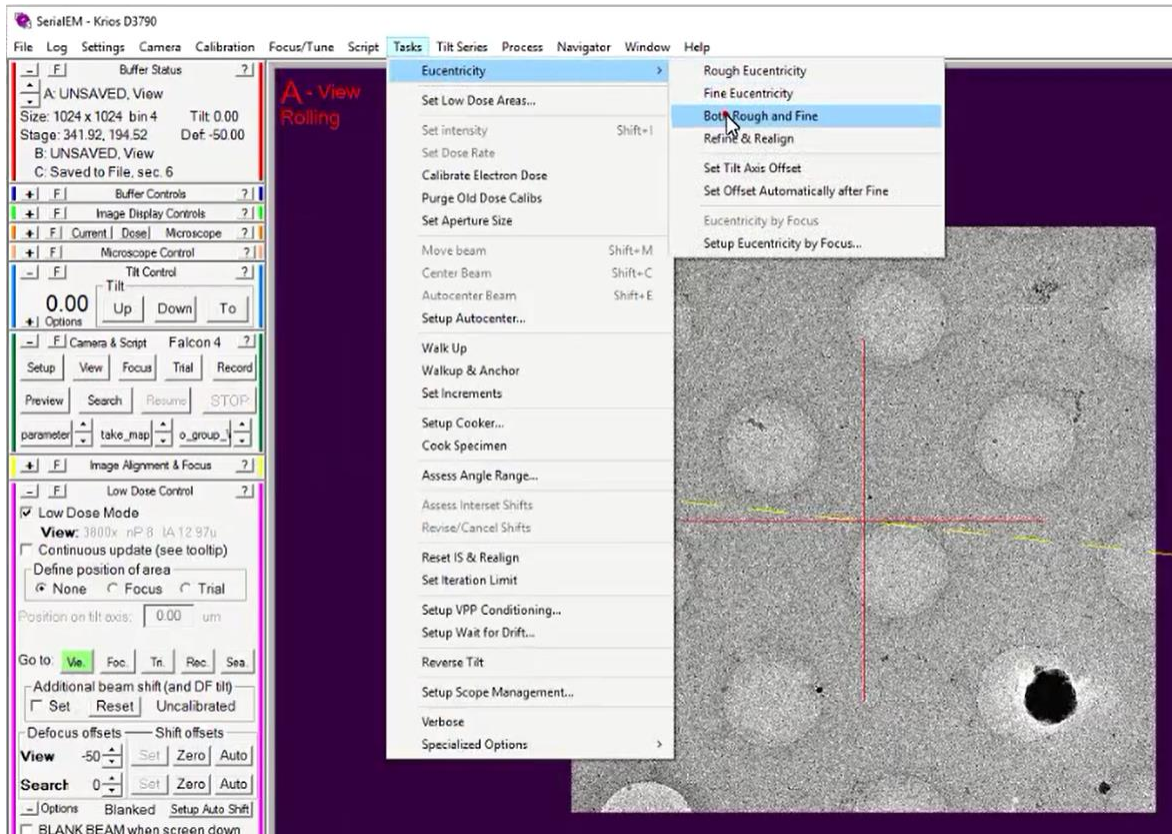
```

3. Select a suitable alignment area:

- For single-particle samples, choose a flat, crack-free carbon film region with minimal ice contamination. For FIB-milled lamellae, if no suitable area is available on the lamella grid, firstly load a carbon-coated grid for alignment, then switch to the lamella grid for mapping and data collection.

4. Adjust Z-height precisely:

- In the top menu, select Tasks → Eccentricity → Both Rough and Fine (Rough $\pm 5^\circ$, Fine $\pm 24^\circ$) to precisely adjust the Z-height.



```

Log
Rough eucentricity: changing Z by -13.48 to 22.52, finished.
Refining eucentricity: Fit to 8 positions gives tilt axis Z displacement
of -1.90 microns and lateral displacement of 0.35 microns

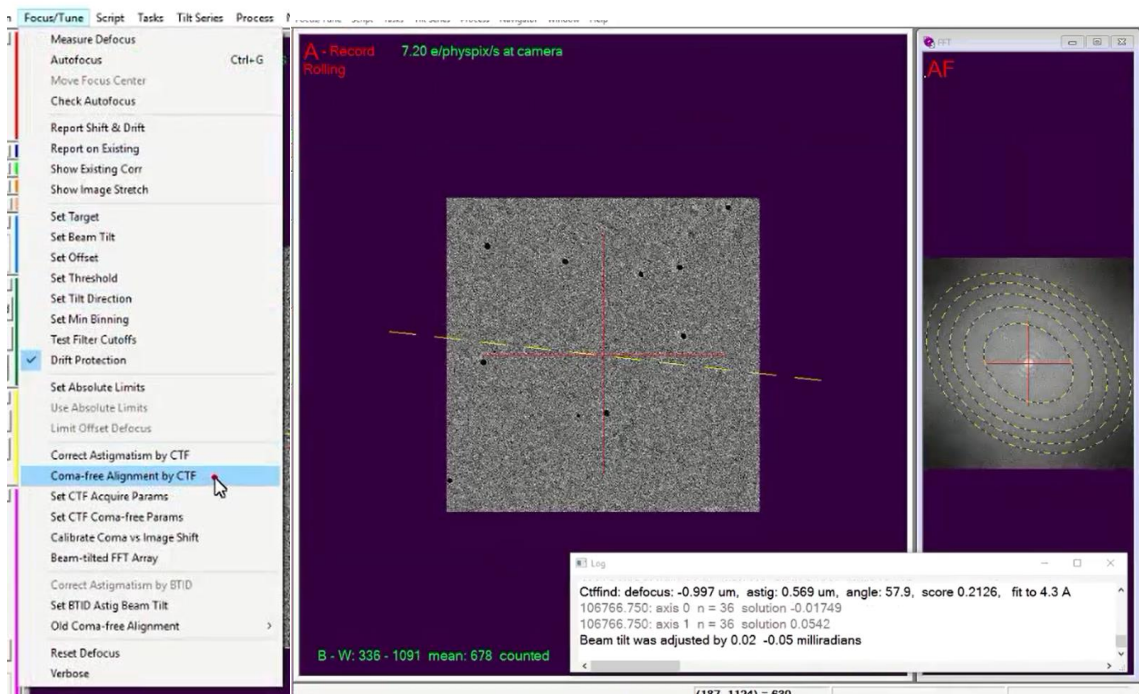
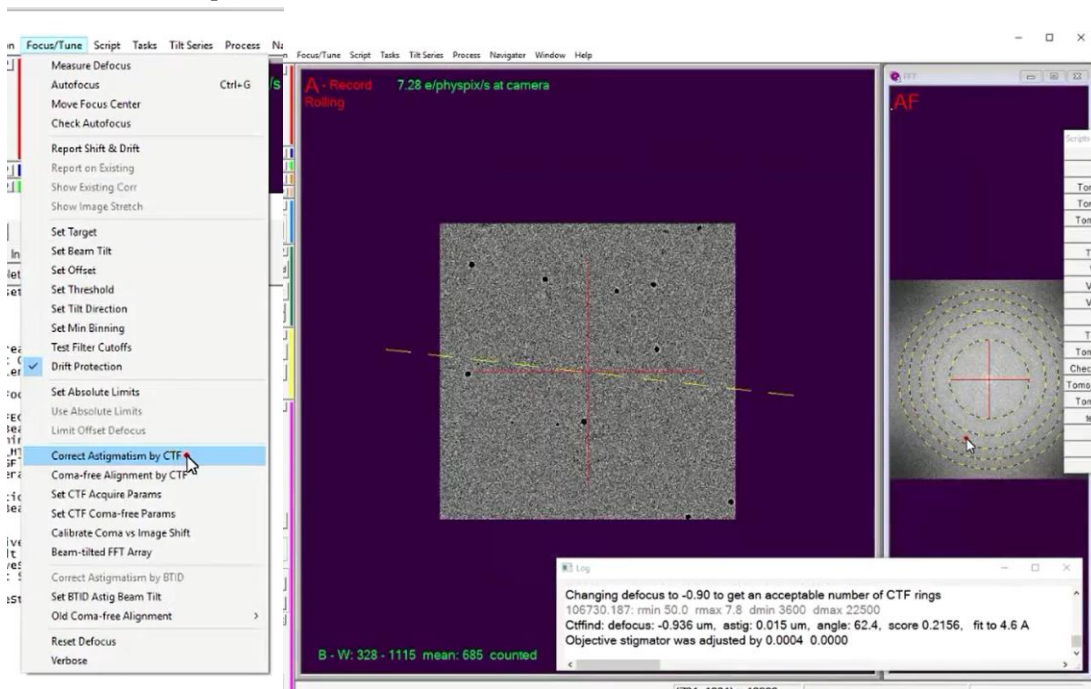
```

- After completion, check the log deviation – it should be within ± 1 (suitable for cryo-ET). If larger, contact service engineering for further

instrument alignment.

5. Correct astigmatism and coma:

- Move to the carbon area and run *Focus/Tune* → *Correct Astigmatism by CTF*, then *Coma-free Alignment by CTF*, and repeat *Correct Astigmatism by CTF*. If the adjustment values in the log are minimal (changing only in the last decimal), alignment is complete; otherwise, repeat these two steps.



6. Save the new parameters in *reset_X*:

- Open *reset_X* again (**keeping the last two lines commented #**), run it, copy the new parameters from the log into those two lines, **remove #**, and run once more.

New alignment parameters

Copy the values into **reset_X** and remove “#” in the last two lines

```
Script &: reset_X
<| Editor >| Up|Script Down|
Indent Fix Increase Reduce|

MacroName reset_X
#130kx
CallMacro 1

GOTOLOWDoseArea R
SetImageShift 0 0
NormalizeAllLenses
Delay 3
SetEucentricFocus

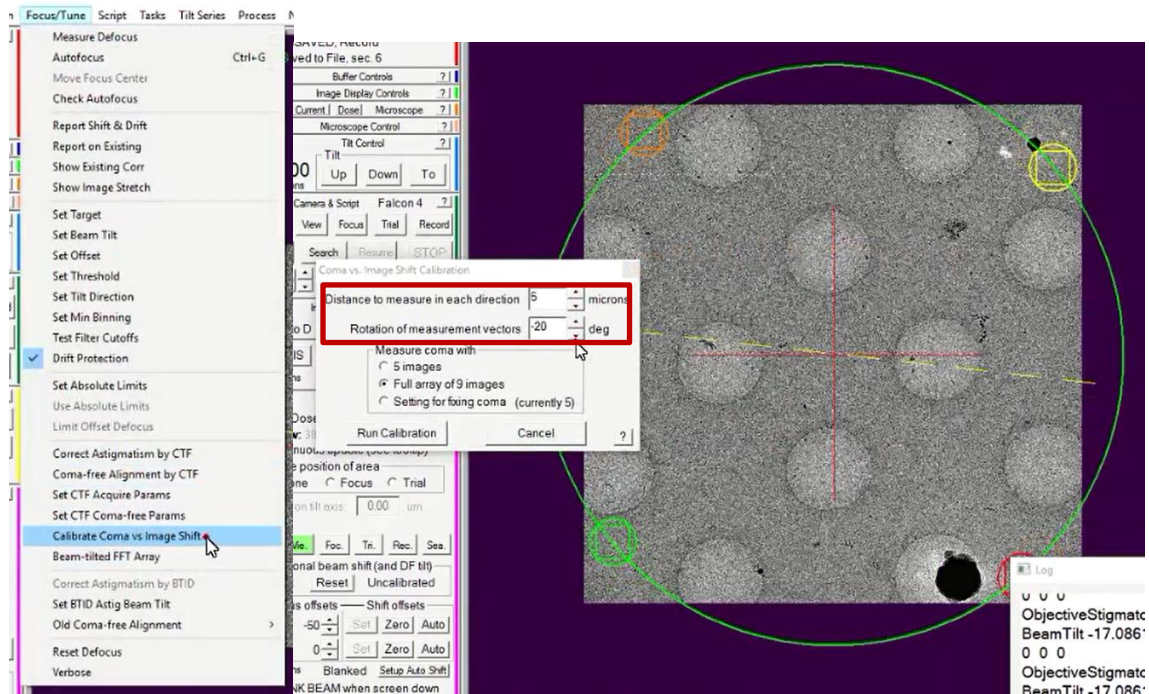
if $Flash_CFEG == 1
  ReportFEGBeamCurrent
  IsFEGFlashingAdvised 1 flash_HT
  if $Flash_HT > 0
    NextFEGFlashHighTemp
    LongOperation FF 0
  endif
  LongOperation FF 1
  ReportFEGBeamCurrent FEG_beam_current
endif

ReportObjectiveStigmator OS_X OS_Y
ReportBeamTilt BT_X BT_Y
echo ObjectiveStigmator $OS_X $OS_Y
echo BeamTilt $BT_X $BT_Y

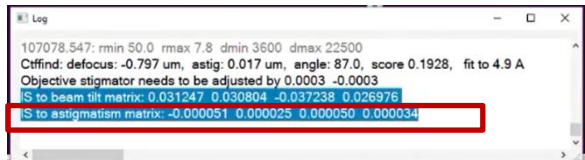
setObjectiveStigmator -0.003321 -0.005128
setBeamTilt -17.086155 -14.75686
```

7. Calibrate Coma vs Image Shift:

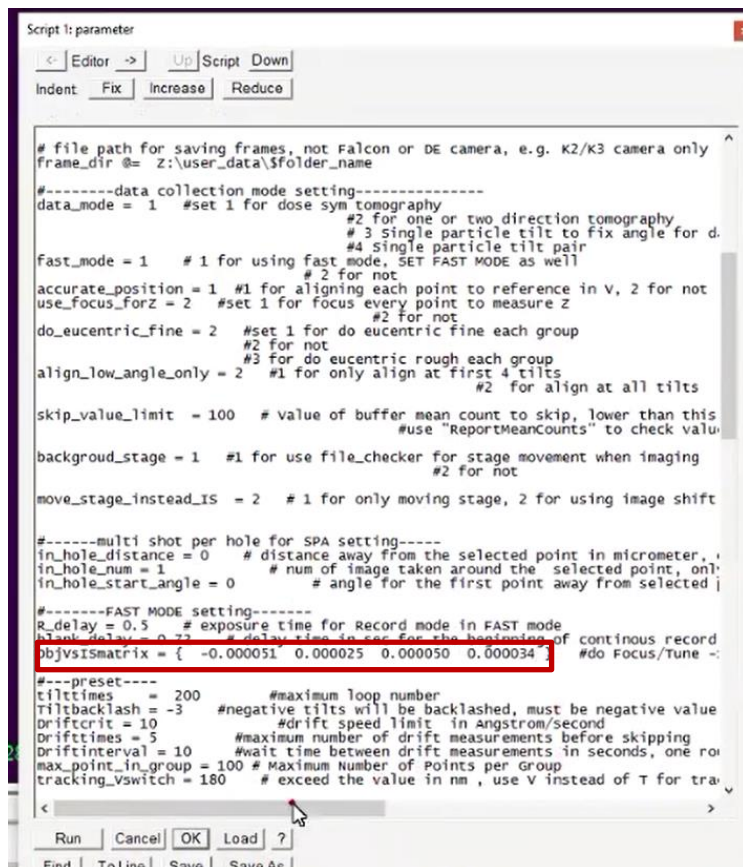
- Under *Focus/Tune* → *Calibrate Coma vs Image Shift*, in View mode, adjust point spacing and angle so that all points are on carbon and free of ice. Click *reset_X* to return to Record magnification, then run the calibration. **Note:** This calibration does not need repeating for every session; once setting saved, it remains valid for the same magnification for up to two weeks.



- Open the parameter script and copy the IS to astigmatism matrix from the log into "objvsiSmatrix"



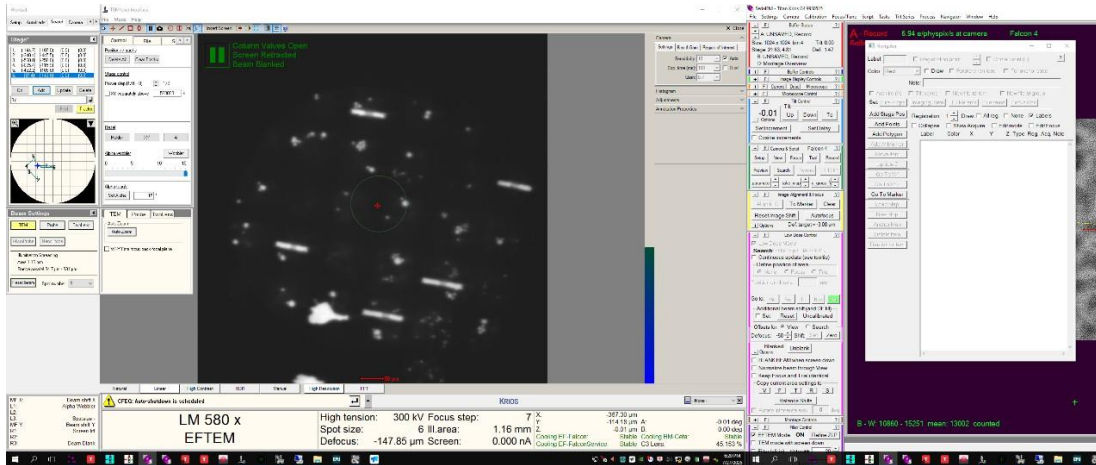
Copy into **parameter** script
objvsiSmatrix



2.4 Low-Magnification Map Acquisition

The procedure for acquiring low-magnification maps in **CR-BIS** is essentially identical to conventional cryo-EM data collection. The steps are as follows:

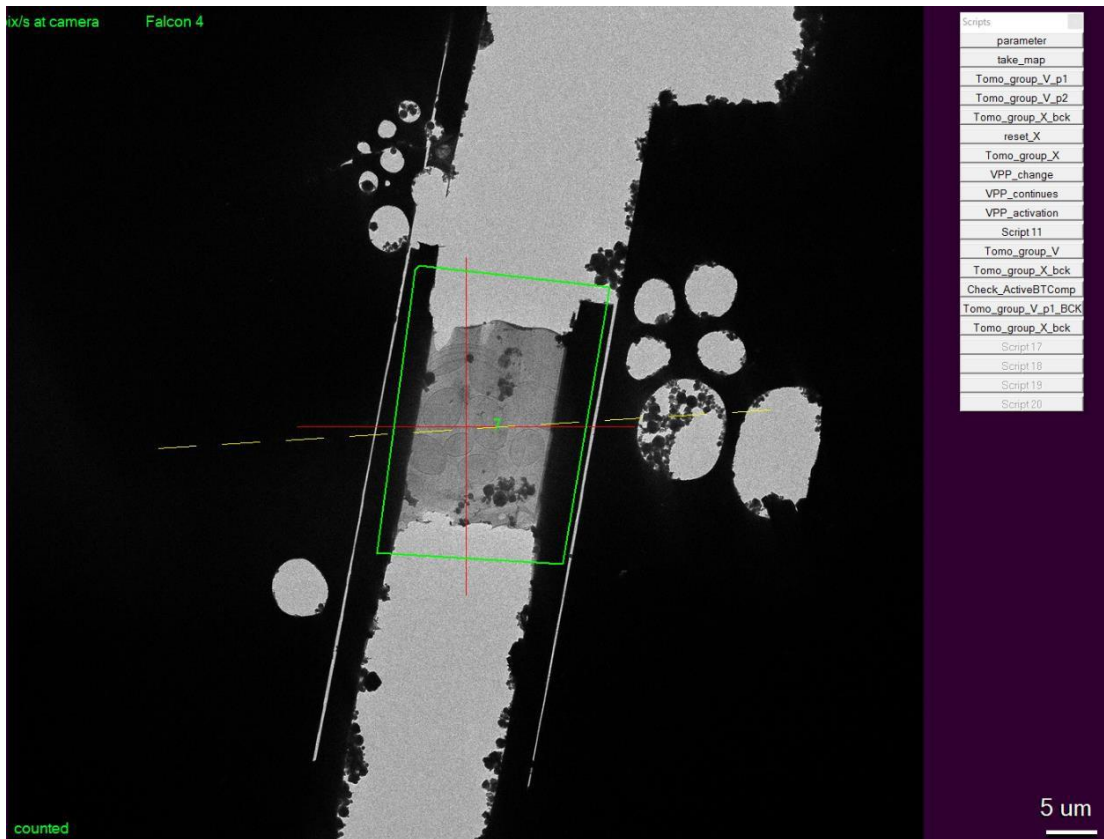
1. At low magnification, locate the target **mesh** or **lamella**.
2. Adjust the Z-height manually or automatically using **Tasks** → **Eucentric** to bring the specimen to the eucentric position.



Low-magnification overview showing mesh or lamella locations. Green boxes indicate areas selected with Navigator → Add Polygon for subsequent high-magnification data collection.

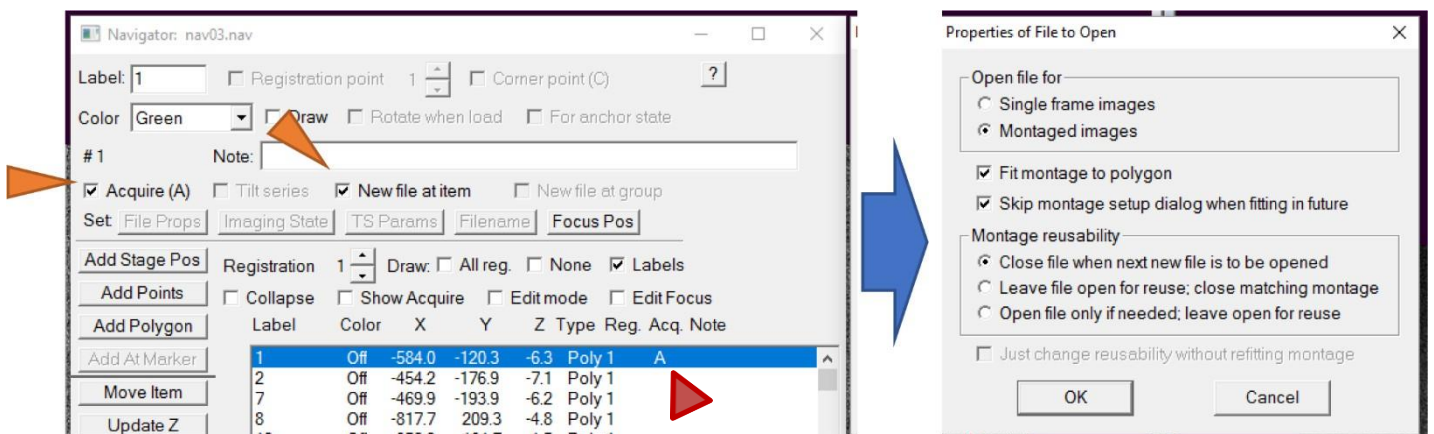
3. In the **Navigator**, outline the desired acquisition regions using **Add Polygon**. Each polygon defines a region (e.g., one lamella or grid square) for subsequent targeting.

Note: When drawing the polygon, ensure its area is slightly larger than your target data collection region. Also, avoid placing target points too close to the polygon edges; otherwise, during the subsequent Tomo_group_V_p1 process, the system may frequently move the stage for low-mag View alignment, which will add unnecessary time overhead.



A representative lamella imaged at low magnification. The green polygon outlines the acquisition area.

4. Switch to **View** mode, and set exposure parameters such as exposure time, binning, and gain reference.
5. Select the polygon(s) for mapping and check the “**Acquire**” (A) box. Specify an output filename (e.g., map01.mrc).



Montage Setup

Camera
☐ Ceta
☒ Falcon 4i

FITTING TO NAVIGATOR AREA: Change mag to adjust number of pieces. Changing mag, binning, overlap, or "Move stage" will refit to area

Magnification: 8700 Binning: 2
Pixel size: 2.89 nm

Number of pieces in X: 4 Y: 9
Piece size in X: 2048 Y: 2048
Overlap in X: 612 Y: 464 Reset

Minimum overlap: 20% and 0.5 micron
Total Area: 6356 x 14720 pixels
18.3 x 42.5 microns Update

☒ Move stage instead of shifting image
☐ Image shift in blocks, IS up to 7 microns

☐ Skip pieces outside Navigator item 0
☐ Do full rectangle; ignore list of pieces to skip
☒ Make map from each montage if Navigator open
☒ Close file when montage is done

☐ Use Montage Mapping, not Record parameters
☒ Use View parameters in Low Dose mode
☐ Use Search parameters in Low Dose mode
☐ Acquire with Multiple Record routine

☐ Use continuous mode with settling factor 0.5
☒ Turn off Drift Correction for stage montage
☐ Use settings for high-quality stage montage

OK Cancel ?

6. Execute Navigator → Acquire at Items to record the low-magnification map.

Acquire at Items

Parameters for: Mapping Final Data

Primary Action
☒ Acquire and save image or montage
☒ Make Navigator map

Primary Action-related
Acquire non-montage images in Low Dose with
☐ Record ☒ View ☐ Search

MONTAGES will be saved into the file:
map07.mrc

Task-related options
General options
☒ Hide unselected options

Tasks before or after Primary (Push button to show controls)
☒ Run Script before Action Every item
Script to run: take_map
Primary Action (Acquire Image) occurs here

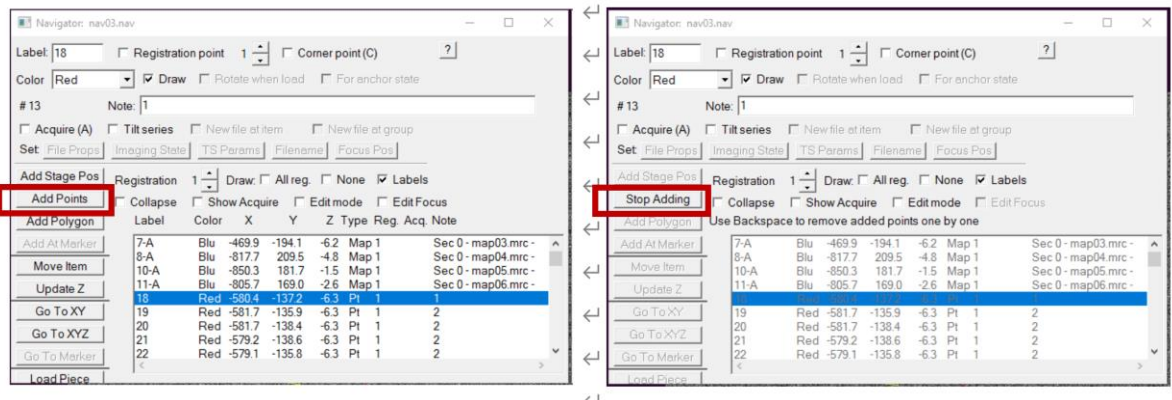
☒ Hide unused tasks Restore Default Order

Save Parameters to File Read in Parameters GO Postpone Cancel ?

2.5 Target Selection

Once maps are acquired, define imaging targets in groups for automated **CR-BIS** collection.

1. Double-click the **map** in the Navigator to open it.
2. Select targets either manually or automatically, organizing them into **groups**. Each group should contain several nearby points.
3. The **first point of each group** serves as the *alignment/focus position* and is **not recorded** as data.
4. The remaining points in the group are acquisition targets. Recommended distances between the first and subsequent points: **Lamella (cryo-ET)**: < 6 μm ; **Single-particle (SPA)**: up to 10 μm (maximum practical distance)
Distances should respect the beam-shift and coma-correction range.



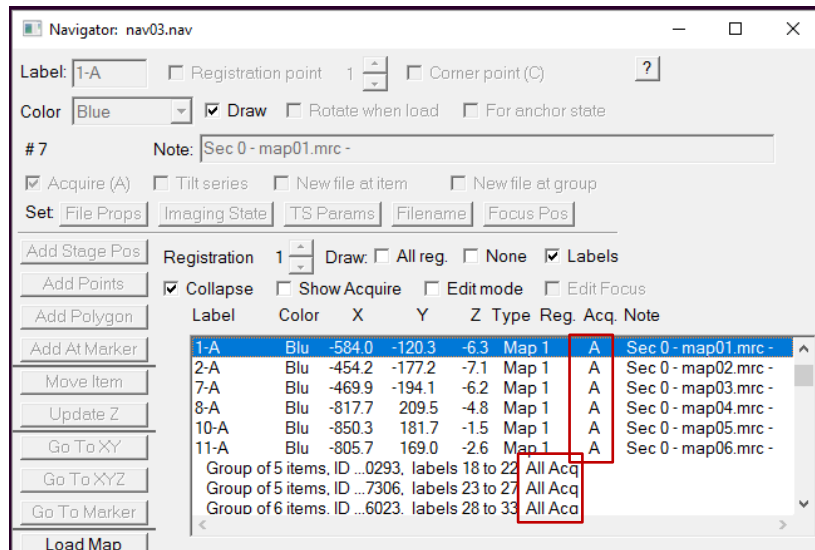
Left: Begin new group selection; **Right:** Complete current group selection

5. Continue defining new groups until all target regions are covered.

2.6 Production Data Acquisition

After all maps and targets are defined, proceed with **CR-BIS data acquisition**:

1. Verify that all maps and points are **checked with "A"** in the Navigator. If polygons also have "A" marks, **remove them** (only points should remain checked).

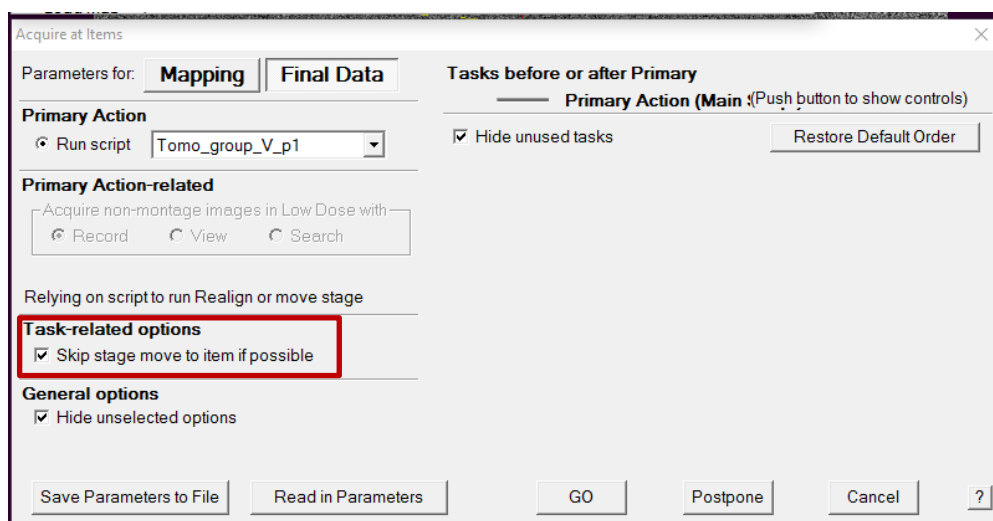


Ensure that each **Map item precedes its corresponding Points** in the Navigator list (maps must appear above their points).

2. Run the pre-alignment script via **Acquire at Items** →

FinalData(Tomo_group_V_p1) to initialize alignment and metadata recording.

If acquisition is interrupted and the order of maps or points changes, **delete map_info.txt** before resuming to avoid indexing errors.



3. Using the file_checker.py script for background stage movement

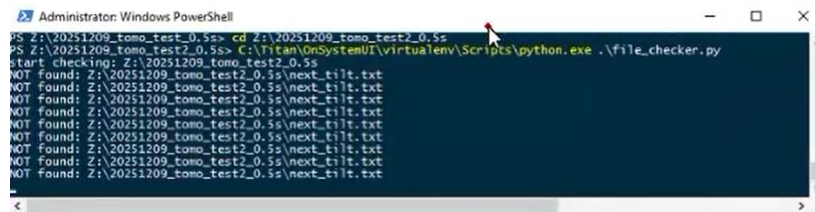
If background_stage = 1 is set in the parameter file, the additional stage-movement script file_checker.py will be used to pre-move the stage for acceleration. Before starting formal data collection, copy file_checker.py to the map storage directory specified in the parameters.

```
background_stage = 1 #1 for use file_checker for stage movement when imaging  
#2 for not
```

```
#-----output data folder setting-----  
#name of folder for saving frames  
folder_name @= 20250727_YtTomo_yq  
  
# map output file path and  
work_dir @= Z:\user_data\%folder_name  
  
# file path for saving frames, not Falcon or DE camera, e.g. K2/K3 camera only  
frame_dir @= Z:\user_data\%folder_name
```

For Falcon cameras, you can use the Thermo Fisher-provided Python interpreter. Example command:

/Titan/OnSystemUI/virtualenv/Scripts/python.exe ./file_checker.py



```
Administrator: Windows PowerShell  
PS Z:\20251209_tomo_test2_0.5s> cd Z:\20251209_tomo_test2_0.5s  
PS Z:\20251209_tomo_test2_0.5s> C:\Titan\OnSystemUI\virtualenv\Scripts\python.exe ./file_checker.py  
start checking: Z:\20251209_tomo_test2_0.5s  
NOT Found: Z:\20251209_tomo_test2_0.5s\next_tilt.txt  
NOT Found: Z:\20251209_tomo_test2_0.5s\next_tilt.txt  
NOT Found: Z:\20251209_tomo_test2_0.5s\next_tilt.txt  
NOT Found: Z:\20251209_tomo_test2_0.5s\next_tilt.txt  
NOT Found: Z:\20251209_tomo_test2_0.5s\next_tilt.txt  
NOT Found: Z:\20251209_tomo_test2_0.5s\next_tilt.txt  
NOT Found: Z:\20251209_tomo_test2_0.5s\next_tilt.txt  
NOT Found: Z:\20251209_tomo_test2_0.5s\next_tilt.txt
```

Once the script is successfully launched, it can be left running in the background.

4. Start the main acquisition run using **Acquire at Items → FinalData (Tomo_group_X)**.

Acquire at Items

Parameters for: **Mapping** **Final Data**

Primary Action

Run script Tomo_group_X

Primary Action-related

Acquire non-montage images in Low Dose with

Record View Search

Relying on script to run Realign or move stage

Task-related options

✓ Skip stage move to item if possible

General options

✓ Hide unselected options

Tasks before or after Primary

Primary Action (Main) (Push button to show controls)

Hide unused tasks Restore Default Order

Save Parameters to File Read in Parameters GO Postpone Cancel ?

5. Monitor the acquisition log to confirm that data writing proceeds normally.

Part 3: Preprocessing of CR-BIS Output Data

All preprocessing scripts require Python 3.

3.1 Processing TIF-format Data

Script: `group_single_point_tif_MotionCorr2_parallel_v2_tiff.py`

Usage:

`python3 group_single_point_tif_MotionCorr2_parallel_v2_tiff.py [options]`

`input_directory`

Generates separated single-point files.

Required: `input_directory` (path to raw CR-BIS data).

Threshold (default 0.1)

```
YQ-desktop:/disk1/github/CR-BIS_preprocess_script/CR-BIS_script_v3/data_process/K3/test_output> ls ../test_data/
K2_00011.tif K2_00012.tif
YQ-desktop:/disk1/github/CR-BIS_preprocess_script/CR-BIS_script_v3/data_process/K3/test_output> python3 ../group_single_point_tif_MotionCorr2_parallel_v2_tiff.py ../test_data/
Processing file: ../test_data/K2_00011.tif
Processing frame 1...
Frame intensity: 0.0054868557042196155
Frame 1 is black.
Processing frame 2...
Frame intensity: 0.06973905723905724
Frame 2 is not black.
Processing frame 3...
Frame intensity: 0.13589316382915173
Frame 3 is not black.
Processing frame 4...
Frame intensity: 0.13579096789467254
Frame 4 is not black.
Processing frame 5...
Frame intensity: 0.1357456240835777
Frame 5 is not black.
Processing frame 6...
Frame intensity: 0.13582497310130878
Frame 6 is not black.
Processing frame 7...
Frame intensity: 0.1358675591601499
Frame 7 is not black.
Processing frame 8...
Frame intensity: 0.1358957943005322
Frame 8 is not black.
Processing frame 9...
Frame intensity: 0.13578652367084826
Frame 9 is not black.
Processing frame 10...
Frame intensity: 0.13581205409267405
Frame 10 is not black.
Processing frame 11...
Frame intensity: 0.13582951278578798
Frame 11 is not black.
Processing frame 12...
Frame intensity: 0.13576342219235366
Frame 12 is not black.
Processing frame 13...
Frame intensity: 0.13563498518450634
Frame 13 is not black.
Processing frame 14...
Frame intensity: 0.13563498518450634
Frame 14 is not black.
Processing frame 15...
Frame intensity: 0.13563498518450634
Frame 15 is not black.
Processing frame 16...
Frame intensity: 0.13563498518450634
Frame 16 is not black.
Processing frame 17...
Frame intensity: 0.13563498518450634
Frame 17 is not black.
Processing frame 18...
Frame intensity: 0.13563498518450634
Frame 18 is not black.
Processing frame 19...
Frame intensity: 0.13563498518450634
Frame 19 is not black.
Processing frame 20...
Frame intensity: 0.13563498518450634
Frame 20 is not black.
Processing frame 21...
Frame intensity: 0.13563498518450634
Frame 21 is not black.
Processing frame 22...
Frame intensity: 0.13563498518450634
Frame 22 is not black.
Processing frame 23...
Frame intensity: 0.13563498518450634
Frame 23 is not black.
Processing frame 24...
Frame intensity: 0.13563498518450634
Frame 24 is not black.
Processing frame 25...
Frame intensity: 0.13563498518450634
Frame 25 is not black.
Processing frame 26...
Frame intensity: 0.13563498518450634
Frame 26 is not black.
Processing frame 27...
Frame intensity: 0.13563498518450634
Frame 27 is not black.
Processing frame 28...
Frame intensity: 0.13563498518450634
Frame 28 is not black.
Processing frame 29...
Frame intensity: 0.13563498518450634
Frame 29 is not black.
Processing frame 30...
Frame intensity: 0.13584238936746498
Frame 30 is not black.
Processing frame 31...
Frame intensity: 0.13587654306845334
Frame 31 is not black.
Processing frame 32...
Frame intensity: 0.11486187691769849
Frame 32 is not black.
Processing frame 33...
Frame intensity: 0.0051727795427392205
Frame 33 is black.
Processing frame 34...
Frame intensity: 0.00514913712053329
Frame 34 is black.
Processing frame 35...
Frame intensity: 0.005154217748723797
Frame 35 is black.
Processing frame 36...
Frame intensity: 0.005130389526990334
Frame 36 is black.
Processing frame 37...
Frame intensity: 0.11269702018504942
Frame 37 is not black.
Processing frame 38...
Frame intensity: 0.13671738173061257
Frame 38 is not black.
Processing frame 39...
Frame intensity: 0.13669564852150537
Frame 39 is not black.
Processing frame 40...
Frame intensity: 0.13676719097901052
Frame 40 is not black.
Processing frame 41...
Frame intensity: 0.13686062574671445
Frame 41 is not black.
Processing frame 42...
Frame intensity: 0.13664720754249485
Frame 42 is not black.
Saved group 1 to K2_00011_1.tif
Processing frame 43...
Frame intensity: 0.13661950273908438
Frame 43 is not black.
```

Red box: unblank frames, blue box: blank frames.

1. Calculate T_{delay} from the number of blank frames multiplied by the frame time. (eg. 4 frames are black, frame time is 0.05s/frame, $T_{\text{delay}} \approx 0.20$ s)
2. Adjust the intensity Threshold according to the measured intensity of the blank frames.

3.2 Processing EER-format Data

```
20260109_170347_EER_GainReference.gain Falcon4_00016.eer
Falcon4_00000.eer Falcon4_00017.eer
Falcon4_00001.eer Falcon4_00018.eer
Falcon4_00002.eer Falcon4_00019.eer
Falcon4_00003.eer Falcon4_00020.eer
Falcon4_00004.eer Falcon4_00021.eer
Falcon4_00005.eer Falcon4_00021.eer_log
Falcon4_00006.eer Falcon4_00022.eer
Falcon4_00007.eer Falcon4_00022.eer_log
Falcon4_00008.eer Falcon4_00023.eer
Falcon4_00009.eer Falcon4_00023.eer_log
Falcon4_00010.eer Falcon4_00024.eer
Falcon4_00011.eer Falcon4_00024.eer_log
Falcon4_00012.eer Falcon4_00025.eer
Falcon4_00013.eer Falcon4_00025.eer_log
Falcon4_00014.eer Falcon4_00026.eer
Falcon4_00015.eer Falcon4_00026.eer_log
Falcon4_00016.eer Falcon4_00027.eer
Falcon4_00017.eer Falcon4_00027.eer_log
```

First 5 tilts
BIS outputs

CR-BIS
outputs

CR-BIS-Tomo Output Folder Example (Before Splitting)

```
cat Falcon4_00021.eer_log
```

```
5.99553 -0.284781 -1.791169 4 0.041321 6 34 R 29471278
5.99553 -1.637413 1.343705 5 0.020546 6 34 R 29471278
5.99553 2.867066 -0.467366 6 0.016017 6 34 R 29471278
5.99553 1.537317 2.590912 7 0.007401 6 34 R 29471278
```

The acquisition parameters for each target position within a CR-BIS output file are recorded in the corresponding *.eer_log file.

Step 1: Calculate blank frame gap before splitting EER files

Copy several CR-BIS output files into a test folder (increasing the number of files generally improves the accuracy of the distribution statistics), and run:

```
python3 calculate_sep_range_python34.py
```

```
Falcon4_00021.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00021.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 60;
min gap between files is 45, max gap is 47;
min frame number in each file is 26, max is 68
Falcon4_00022.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00022.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 60;
min gap between files is 47, max gap is 48;
min frame number in each file is 23, max is 69
Falcon4_00023.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00023.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 60;
min gap between files is 46, max gap is 47;
min frame number in each file is 26, max is 67
Falcon4_00024.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00024.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 61;
min gap between files is 45, max gap is 46;
min frame number in each file is 25, max is 68
```

```
Falcon4_00054.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00054.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 63;
min gap between files is 47, max gap is 48;
min frame number in each file is 21, max is 68
first_range:(401, 401)
blank_range:(44, 52)
end_range:(53, 71)
```

The statistics of all files in the current directory

For each CR-BIS output EER file, the calculated blank frame gaps will be written into *_json_dose_sep.

```
cat Falcon4_00021_json_dose_sep
```

```
{
  "starts": [
    359,
    472,
    586,
    700
  ],
  "ends": [
    428,
    541,
    655,
    769
  ]
}
```

Step 2: EER file splitting

Modify the split script **eer_writer_5_v4_py34.py**. Based on the results obtained in **Step 1**, adjust the range parameter in the script to a value marginally exceeding the **Step 1** result.

```
if __name__ == "__main__":
    #Falcon4i
    first_range = (380, 500)
    blank_range = (35, 55)
    end_range = (40, 85)
```

Then run:

```
python3 eer_writer_5_v4_py34.py
```

```
usage: eer_writer_5_v4_py34.py [-h] frame_num
eer_writer_5_v4_py34.py: error: the following arguments are required: frame_num
```

The **Frame_num** parameter should be set to the EER frame number specified during data collection; this will determine the frame count of the split output files.

```

20260109_170347_EER_GainReference.gain
calculate_sep_range_python34.py
eer_name.lst
eer_writer_5_v4_py34.py
failed_Falcon4_00092.eer
failed_Falcon4_00144.eer
failed_Falcon4_00584.eer
Falcon4_00000.eer
Falcon4_00001.eer
Falcon4_00002.eer
Falcon4_00003.eer
Falcon4_00004.eer
Falcon4_00005.eer
Falcon4_00006.eer
Falcon4_00007.eer
Falcon4_00008.eer
Falcon4_00009.eer
Falcon4_00010.eer
Falcon4_00011.eer
Falcon4_00012.eer
Falcon4_00013.eer
Falcon4_00014.eer
Falcon4_00015.eer
Falcon4_00016.eer
Falcon4_00017.eer
Falcon4_00018.eer
Falcon4_00019.eer
Falcon4_00020.eer
Falcon4_00021_00.eer
Falcon4_00021_01.eer
Falcon4_00021_02.eer
Falcon4_00021_03.eer
Falcon4_00021_json
Falcon4_00021_json_dose_sep
Falcon4_00021log_sep.txt
Falcon4_00022_00.eer
Falcon4_00022_01.eer
Falcon4_00022_02.eer
Falcon4_00022_03.eer
Falcon4_00022_json
Falcon4_00022_json_dose_sep
Falcon4_00022log_sep.txt

```

Note on output files:

- BIS output files remain unchanged.
- CR-BIS files are split into individual files for each target position, and **the original files are deleted.**
- The *.eer_log file is replaced with ***log_sep.txt**, which will be used for subsequent .mdoc file generation.

```
cat Falcon4_00021log_sep.txt
```

```

5.99553 Falcon4_00021_00.eer -0.284781 -1.791169 4 0.041321 6 34 R 29471278
5.99553 Falcon4_00021_01.eer -1.637413 1.343705 5 0.020546 6 34 R 29471278
5.99553 Falcon4_00021_02.eer 2.867066 -0.467366 6 0.016017 6 34 R 29471278
5.99553 Falcon4_00021_03.eer 1.537317 2.590912 7 0.007401 6 34 R 29471278

```

CR-BIS-Tomo Output Folder Example (After Splitting)

3.3 Cryo-ET mdoc File Generation

During data acquisition, collection parameters are written into a **tomo_log.txt** file.

An example of the original **tomo_log.txt** is shown below:

```
(base) yq@IBPserver192:/data42/YQ/Falcon_data/20260113_fast_tomo/raw_eer$ more tomo_log.txt
tomo_start at Nav_Item 3
-0.004235 Falcon4_00001.eer -0.287855 -1.964547 4 0.043952 1 39 R 29471278
-0.004235 Falcon4_00002.eer -1.649436 1.189066 5 0.042708 1 39 R 29471278
-0.004235 Falcon4_00003.eer 2.857132 -0.626617 6 -0.097707 1 39 R 29471278
-0.004235 Falcon4_00004.eer 1.5175 2.452106 7 -0.13598 1 39 R 29471278
1.99552 Falcon4_00005.eer -0.335217 -1.827753 4 -0 2 38 R 29471278
1.99552 Falcon4_00006.eer -1.695808 1.323987 5 -0 2 38 R 29471278
1.99552 Falcon4_00007.eer 2.81185 -0.493756 6 -0 2 38 R 29471278
1.99552 Falcon4_00008.eer 1.473748 2.582073 7 -0 2 38 R 29471278
-2.00399 Falcon4_00009.eer -0.302623 -1.8338 4 0.075467 3 37 R 29471278
-2.00399 Falcon4_00010.eer -1.663199 1.317912 5 0.078269 3 37 R 29471278
-2.00399 Falcon4_00011.eer 2.839884 -0.49118 6 0.059109 3 37 R 29471278
-2.00399 Falcon4_00012.eer 1.500688 2.586718 7 0.063084 3 37 R 29471278
-4.004744 Falcon4_00013.eer -0.290261 -1.829762 4 0.049585 4 36 R 29471278
-4.004744 Falcon4_00014.eer -1.648848 1.3178 5 0.023153 4 36 R 29471278
-4.004744 Falcon4_00015.eer 2.849413 -0.48091 6 0.030499 4 36 R 29471278
-4.004744 Falcon4_00016.eer 1.512214 2.592982 7 0.022223 4 36 R 29471278
3.995775 Falcon4_00017.eer -0.281635 -1.800779 4 0.037872 5 35 R 29471278
3.995775 Falcon4_00018.eer -1.639578 1.345347 5 0.039278 5 35 R 29471278
3.995775 Falcon4_00019.eer 2.867881 -0.470791 6 0.029663 5 35 R 29471278
3.995775 Falcon4_00020.eer 1.533086 2.598611 7 0.031658 5 35 R 29471278
5.99553 Falcon4_00021.eer 1.537317 2.590912 3 7 6 35 R 29471278
-6.004 Falcon4_00022.eer 1.528771 2.585431 3 7 7 34 R 29471278
-8.004254 Falcon4_00023.eer 1.5329 2.587393 3 7 8 33 R 29471278
7.994785 Falcon4_00024.eer 1.562625 2.599155 3 7 9 32 R 29471278
9.99504 Falcon4_00025.eer 1.571911 2.579486 3 7 10 31 R 29471278
-10.004009 Falcon4_00026.eer 1.550648 2.572743 3 7 11 30 R 29471278
-12.004264 Falcon4_00027.eer 1.551012 2.559123 3 7 12 29 R 29471278
11.993795 Falcon4_00028.eer 1.59549 2.571375 3 7 13 28 R 29471278
13.99355 Falcon4_00029.eer 1.592945 2.568637 3 7 14 27 R 29471278
-14.004519 Falcon4_00030.eer 1.577725 2.518067 3 7 15 26 R 29471278
-16.005274 Falcon4_00031.eer 1.57464 2.49873 3 7 16 25 R 29471278
15.995305 Falcon4_00032.eer 1.623733 2.541011 3 7 17 24 R 29471278
17.99506 Falcon4_00033.eer 1.623813 2.52757 3 7 18 23 R 29471278
-18.005029 Falcon4_00034.eer 1.588438 2.476088 3 7 19 22 R 29471278
-20.005284 Falcon4_00035.eer 1.581466 2.467673 3 7 20 21 R 29471278
19.992816 Falcon4_00036.eer 1.64729 2.500581 3 7 21 20 R 29471278
21.99307 Falcon4_00037.eer 1.646323 2.477845 3 7 22 19 R 29471278
```

For EER-format data:

Run the script **tomo_log_edit.py**, which reads the updated information from the corresponding ***log_sep.txt** file (generated after splitting) and modifies **tomo_log.txt** accordingly. The output is saved as **tomo_log_new.txt**, with a format illustrated below:

```
(base) yq@IBPserver192:/data42/YQ/Falcon_data/20260113_fast_tomo/raw_eer$ more tomo_log_new.txt
tomo_start at Nav_Item 3
-0.004235 Falcon4_00001.eer -0.287855 -1.964547 4 0.043952 1 39 R 29471278
-0.004235 Falcon4_00002.eer -1.649436 1.189066 5 0.042708 1 39 R 29471278
-0.004235 Falcon4_00003.eer 2.857132 -0.626617 6 -0.097707 1 39 R 29471278
-0.004235 Falcon4_00004.eer 1.5175 2.452106 7 -0.13598 1 39 R 29471278
1.99552 Falcon4_00005.eer -0.335217 -1.827753 4 -0 2 38 R 29471278
1.99552 Falcon4_00006.eer -1.695808 1.323987 5 -0 2 38 R 29471278
1.99552 Falcon4_00007.eer 2.81185 -0.493756 6 -0 2 38 R 29471278
1.99552 Falcon4_00008.eer 1.473748 2.582073 7 -0 2 38 R 29471278
-2.00399 Falcon4_00009.eer -0.302623 -1.8338 4 0.075467 3 37 R 29471278
-2.00399 Falcon4_00010.eer -1.663199 1.317912 5 0.078269 3 37 R 29471278
-2.00399 Falcon4_00011.eer 2.839884 -0.49118 6 0.059109 3 37 R 29471278
-2.00399 Falcon4_00012.eer 1.500688 2.586718 7 0.063084 3 37 R 29471278
-4.004744 Falcon4_00013.eer -0.290261 -1.829762 4 0.049585 4 36 R 29471278
-4.004744 Falcon4_00014.eer -1.648848 1.3178 5 0.023153 4 36 R 29471278
-4.004744 Falcon4_00015.eer 2.849413 -0.48091 6 0.030499 4 36 R 29471278
-4.004744 Falcon4_00016.eer 1.512214 2.592982 7 0.022223 4 36 R 29471278
3.995775 Falcon4_00017.eer -0.281635 -1.800779 4 0.037872 5 35 R 29471278
3.995775 Falcon4_00018.eer -1.639578 1.345347 5 0.039278 5 35 R 29471278
3.995775 Falcon4_00019.eer 2.867881 -0.470791 6 0.029663 5 35 R 29471278
3.995775 Falcon4_00020.eer 1.533086 2.598611 7 0.031658 5 35 R 29471278
5.99553 Falcon4_00021_00.eer -0.284781 -1.791169 4 0.041321 6 34 R 29471278
5.99553 Falcon4_00021_01.eer -1.637413 1.343705 5 0.020546 6 34 R 29471278
5.99553 Falcon4_00021_02.eer 2.867066 -0.467366 6 0.016017 6 34 R 29471278
5.99553 Falcon4_00021_03.eer 1.537317 2.590912 7 0.007401 6 34 R 29471278
-6.004 Falcon4_00022_00.eer -0.273592 -1.834182 4 0.001281 7 33 R 29471278
-6.004 Falcon4_00022_01.eer -1.628307 1.304378 5 0.002839 7 33 R 29471278
-6.004 Falcon4_00022_02.eer 2.86223 -0.480364 6 0.004551 7 33 R 29471278
-6.004 Falcon4_00022_03.eer 1.528771 2.585431 7 0.005535 7 33 R 29471278
-8.004254 Falcon4_00023_00.eer -0.272381 -1.826702 4 0 8 32 R 29471278
-8.004254 Falcon4_00023_01.eer -1.620628 1.299627 5 0 8 32 R 29471278
-8.004254 Falcon4_00023_02.eer 2.860319 -0.466981 6 0 8 32 R 29471278
-8.004254 Falcon4_00023_03.eer 1.5329 2.587393 7 0 8 32 R 29471278
7.994785 Falcon4_00024_00.eer -0.268956 -1.764994 4 -0 9 31 R 29471278
7.994785 Falcon4_00024_01.eer -1.614469 1.356418 5 -0 9 31 R 29471278
7.994785 Falcon4_00024_02.eer 2.885041 -0.445256 6 -0 9 31 R 29471278
7.994785 Falcon4_00024_03.eer 1.562625 2.599155 7 -0 9 31 R 29471278
```

In this new file, the CR-BIS-related file information is replaced with the paths and details of the split output files. Subsequently, run **mdoc.py** to generate the **.mdoc** file for each tilt series.



For TIFF-format data:

First, execute **modi_tif_log.py** to create an updated ***modi.tif_log** file. Then, run **tomo_log_edit.py** to modify **tomo_log.txt** using the updated log. Finally, use **mdoc.py** to generate the **.mdoc** files for each tilt series.